TECHNICAL DOCUMENTATION & ENGINEERING STANDARDS

Enterprise REST API Standards & Design Guidelines

URI Schema Conventions, HTTP Verb Semantics, Error Envelopes, and Throttling

Standard: RESTful / OpenAPI 3.0 **Version:** 4.1 **Owner:** API Governance Council **Source Reference:** Django REST Framework Docs

1. URI Structure and Resource Naming Conventions

Apex RESTful endpoints represent resources using plural, lowercase nouns. URIs must never contain actions or verbs:

- **Correct:** `GET /api/v1/documents` (List documents)
- **Correct:** `POST /api/v1/documents` (Create document)
- **Correct:** `GET /api/v1/documents/{id}/chunks` (List chunks belonging to document)
- **Forbidden:** `POST /api/v1/getDocument` or `POST /api/v1/save-doc`

2. HTTP Method Semantics and Status Code Matrix

Method	Semantics	Idempotent?	Standard HTTP Status Codes
GET	Retrieve resource	Yes	`200 OK`, `404 Not Found`
POST	Create resource	No	`201 Created`, `400 Bad Request`, `422 Unprocessable`
PUT	Full replacement	Yes	`200 OK`, `400 Bad Request`, `404 Not Found`
PATCH	Partial update	Yes	`200 OK`, `400 Bad Request`, `404 Not Found`
DELETE	Remove resource	Yes	`204 No Content`, `404 Not Found`

3. Standard Error Envelope Schema

All 4xx and 5xx API error responses must adhere strictly to the standardized enterprise JSON envelope:

```
{
  "status": "error",
  "error": {
    "code": "VALIDATION_ERROR",
    "message": "The submitted payload failed schema validation.",
    "details": [
      {
        "field": "query",
        "issue": "Field is required and cannot be null."
      }
    ]
  },
  "timestamp": "2026-10-17T08:00:00Z",
  "request_id": "req-98f2b34a-91dc-4a11"
}
```

4. Pagination, Versioning, and Rate Limiting

- **Cursor-Based Pagination:** Standard for real-time querying: `/api/v1/query?limit=20&cursor=cD0yMDI2`. Returns `next` and `previous` cursor links.
- **API Versioning:** Explicit URI path versioning (e.g., `/v1/`, `/v2/`). Deprecated versions are maintained for 12 months with `Deprecation` and `Sunset` HTTP headers.
- **Throttling:** Enforce rate limiting returning standard headers: `X-RateLimit-Limit: 1000`, `X-RateLimit-Remaining: 842`, `X-RateLimit-Reset: 1776412800`. Excess requests return `429 Too Many Requests`.